DAgger and HG-DAgger

HG-DAgger: Interactive Imitation Learning with Human Experts

Michael Kelly, Chelsea Sidrane, Katherine Driggs-Campbell, Mykel J. Kochenderfer · Stanford / UIUC · ICRA 2019

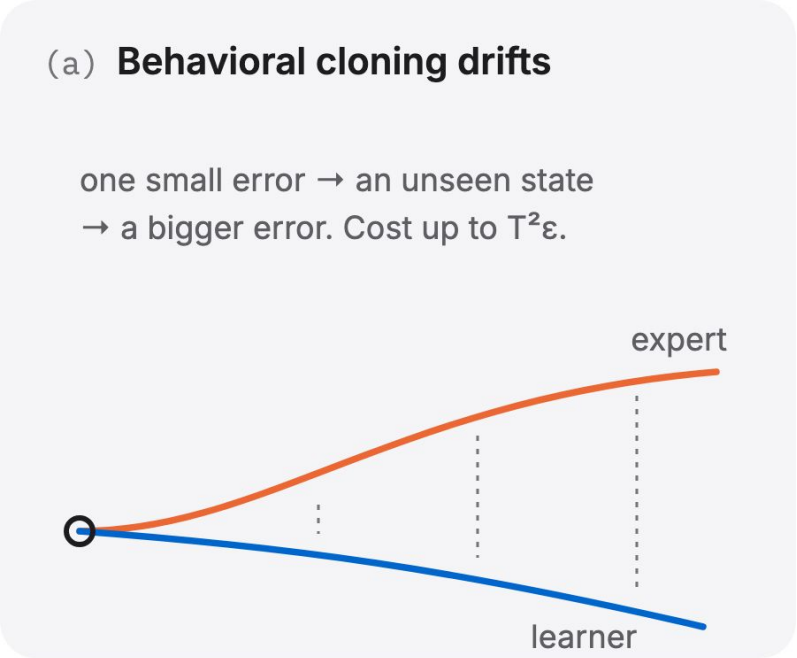


A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

Stéphane Ross, Geoffrey J. Gordon, J. Andrew Bagnell · CMU · AISTATS 2011

Pavel Bushuyeu · University of Hawai‘i at Mānoa · November 2026

Errors change the states the robot visits



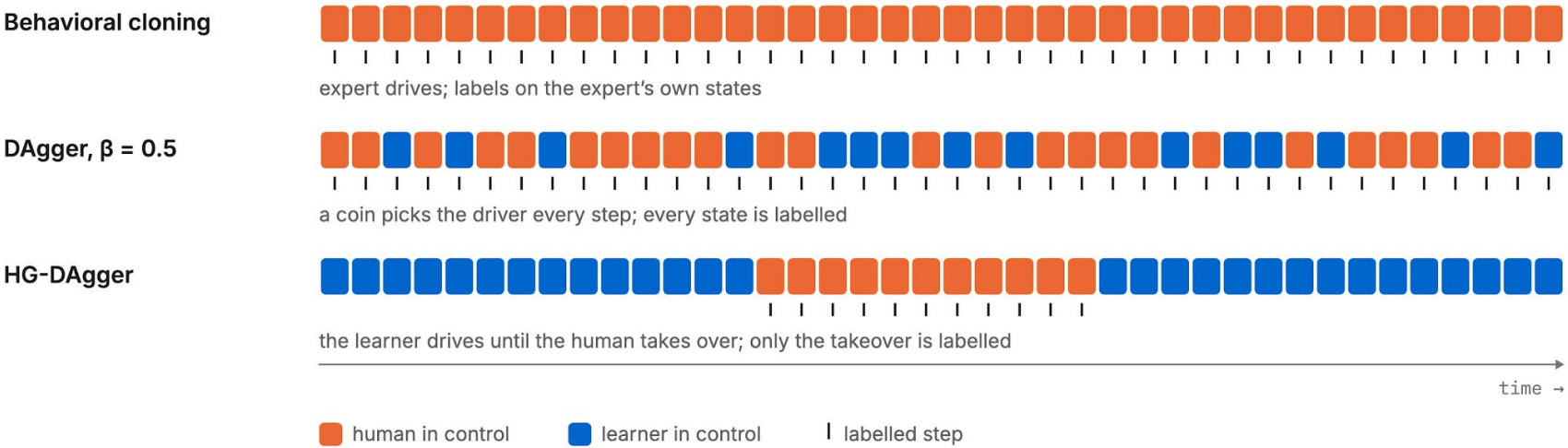
Behavioral cloning learns from expert demonstrations.

Its own mistakes lead to unfamiliar states.

Worst-case excess cost $T^2\varepsilon$

T = episode length; ε = imitation error on expert states

The same training loop, different control



Roll out → label → aggregate → retrain → repeat

A data guarantee and a learned risk threshold

DAgger

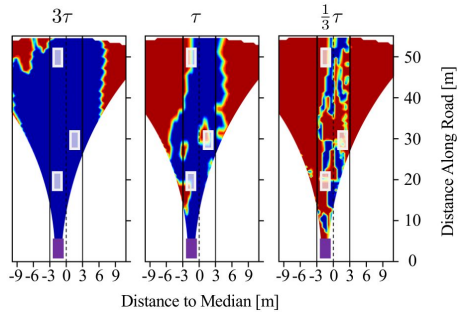
No-regret learning on aggregated data

Cost bound of order Tε under the paper’s assumptions

A stationary policy selected from the training sequence

HG-DAgger

Full human control
Doubt = ensemble disagreement
Threshold learned at takeovers

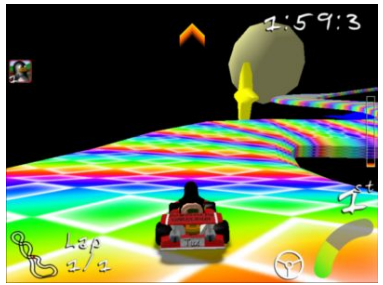


Kelly et al., Fig. 6: safe / unsafe estimates

Games and obstacle avoidance

DAgger: Super Tux Kart

0 falls per lap
after 15 training rounds



Ross et al., Section 5.1 and Fig. 2

HG-DAgger: physical car

Method	Hits	Off-road
Cloning	1	6
DAgger	1	1
HG-DAgger	0	0

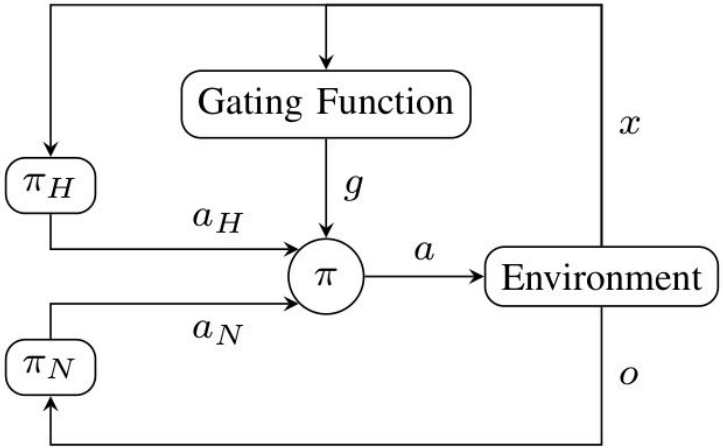
First 5,000 samples; physical car
with simulated obstacles. Kelly et al., Table II.

Human control affects the training labels

Collect corrections on learner states.

Give the expert full authority while correcting.

Use takeover events as evidence about risk.



For XR: the takeover interface is part of data collection.

Limits of the evidence

DAgger

Expert labels at every visited state

Repeated collection and retraining

Unsafe learner rollouts on real hardware

HG-DAgger

No controlled user study of lag or authority

Small physical-car evaluation

Human must detect danger soon enough

Risk threshold remains a heuristic

Control authority and latency in VR

How do takeover design and delay affect useful corrections?

Proposed study

Full takeover vs shared control

Low vs added latency

Same recorded failure starts

Measures

Recovery success and time

Action-label consistency

Operator workload

AI use: drafting and slide construction; claims checked against both papers.